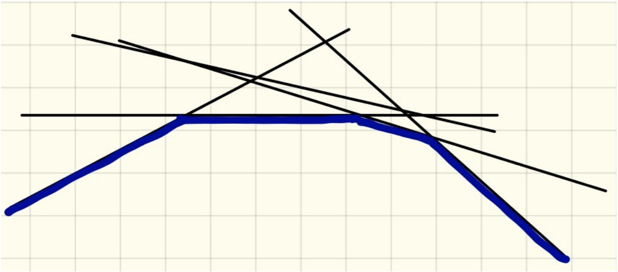
Basics of Convex Optimization

Dual MAP LP Objective:

Dual:

$$\max_{\phi \in \mathbb{R}^J} \sum_{v \in \mathcal{V}} \min_{s \in \mathcal{X}_v} \theta_v^{\phi}(s) + \sum_{uv \in \mathcal{E}} \min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^{\phi}(s,t)$$



- concave
- unconstrained
- non-smooth, piece-wise linear

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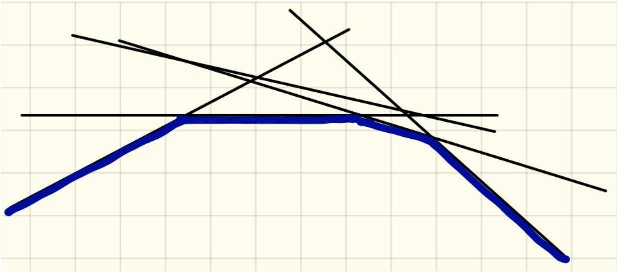

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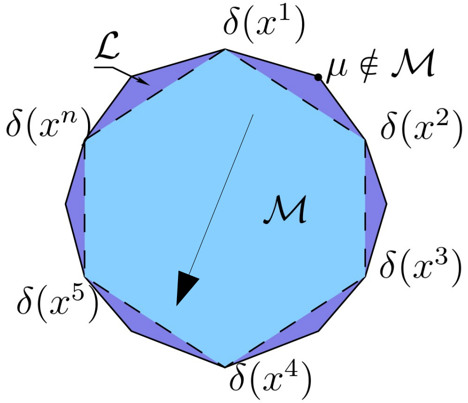
Primal: $\min_{\mu\in\mathcal{L}} \langle\theta,\mu\rangle$



- convex
- constrained
- non-smooth, LP

Dual MAP LP Objective: Number of Variables

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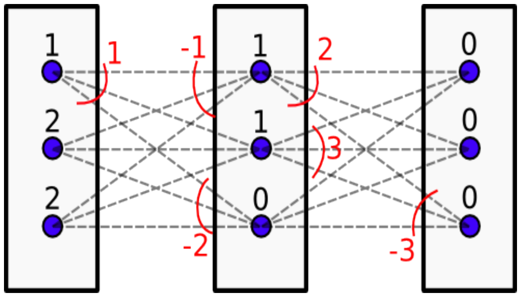
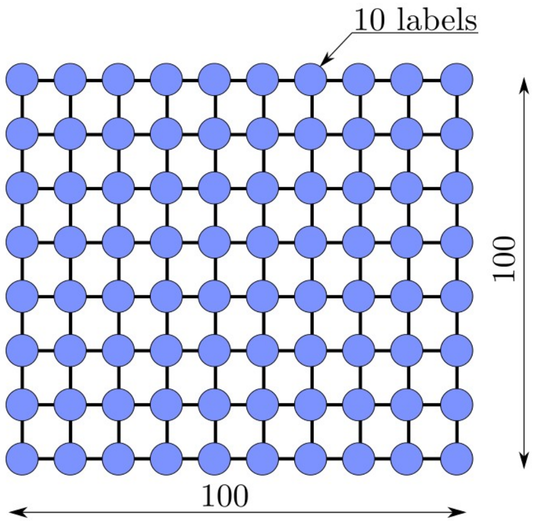


1) 10^5 and less
2) 10^6
3) 10^7
4) 10^8
5) 10^9
6) 10^{10} and more



?

Pascal VOC

$500 \times 300 \times 21$

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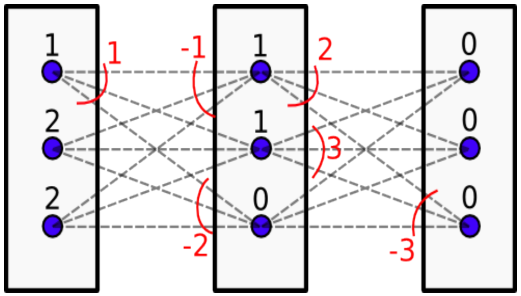

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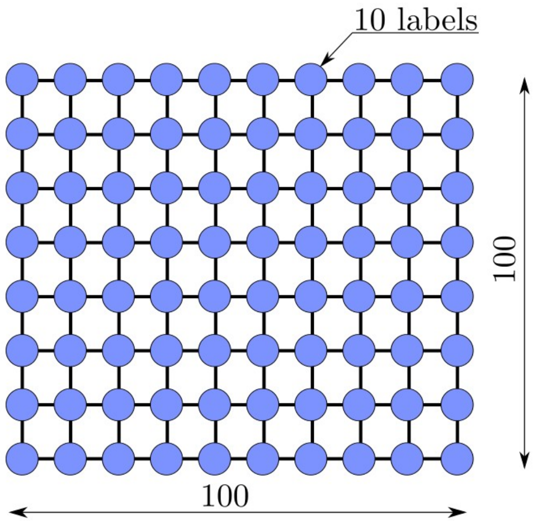

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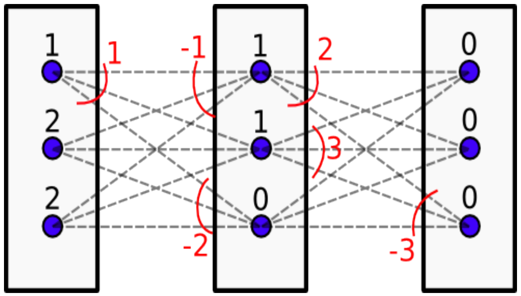
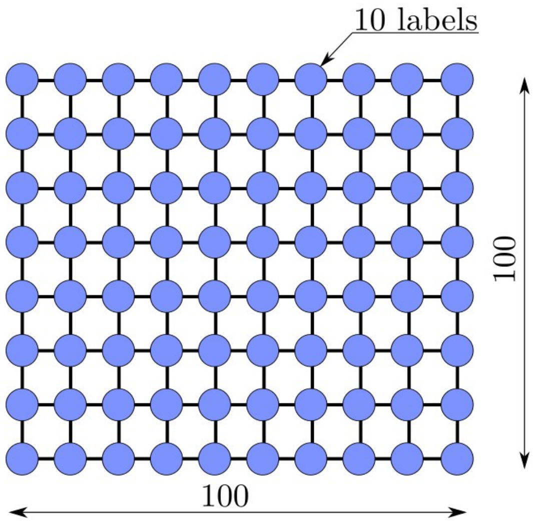


Dual is preferable to optimize.



primal 10^6, dual 10^5

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Primal 10^9, dual 10^7

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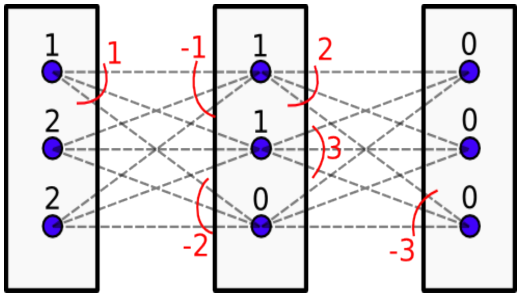
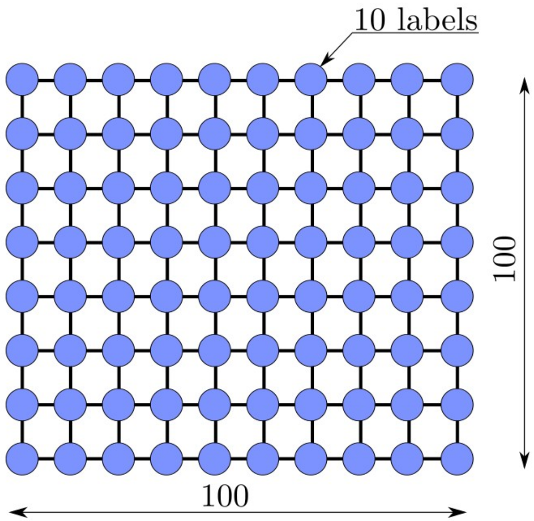


Large-scale problems,
we should stick to first order methods!



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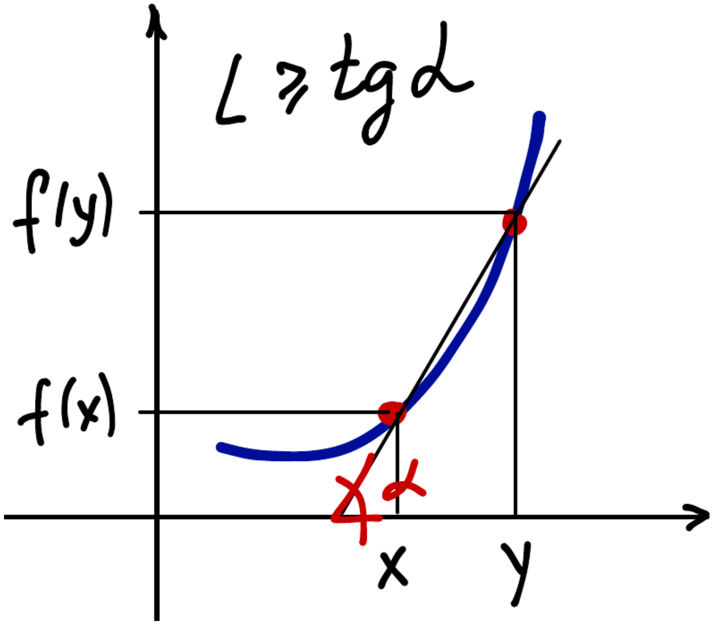
Definition: *f* is Lipschitz-continuous on *D* if for some L>=0 it holds

$$|f(x) - f(y)| \leq L\|x - y\| \quad \forall x, y \in D$$

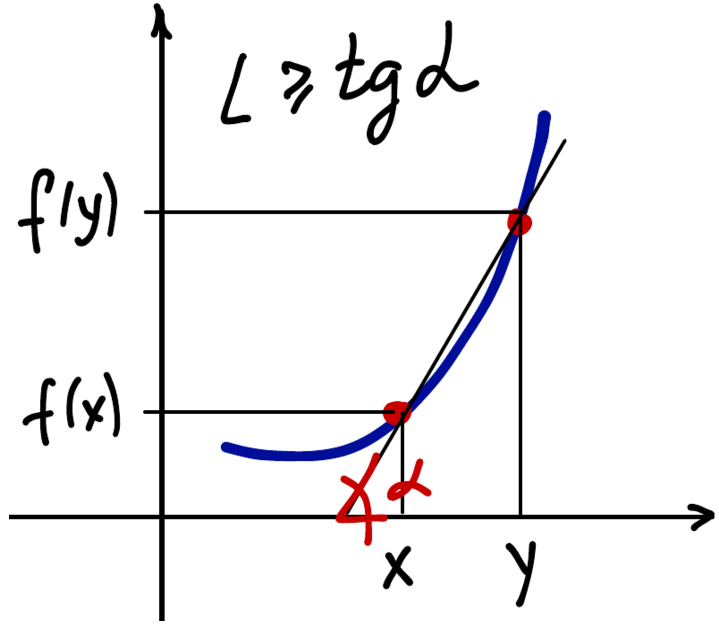


Proposition: *f* differentiable $\Rightarrow$ $L \geq \sup_{x \in D} |\nabla f(x)|$

Is *f* Lipschitz-continuous on *D*? If yes, what is the min. value of *L*?

$$f(x) = x, \quad D = \mathbb{R}$$

1) Yes, 0
2) Yes, 1
3) Yes, 2
4) No
5) None is correct
6) I do not know



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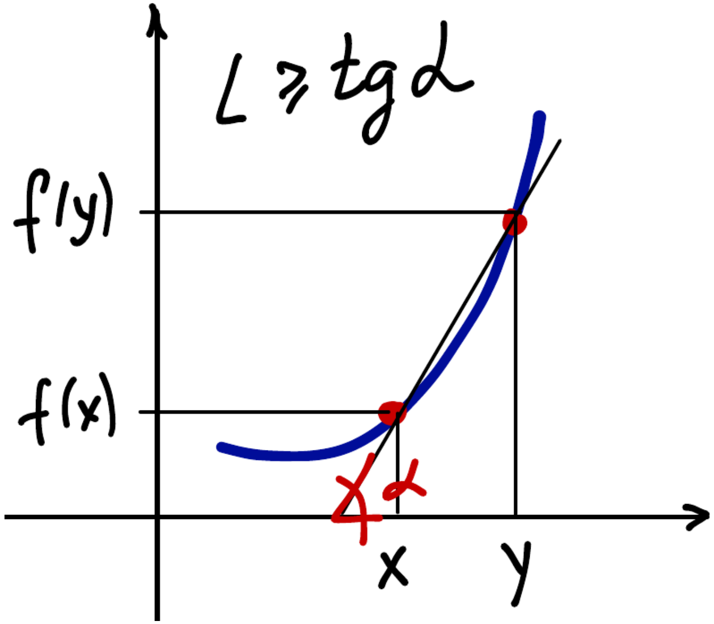

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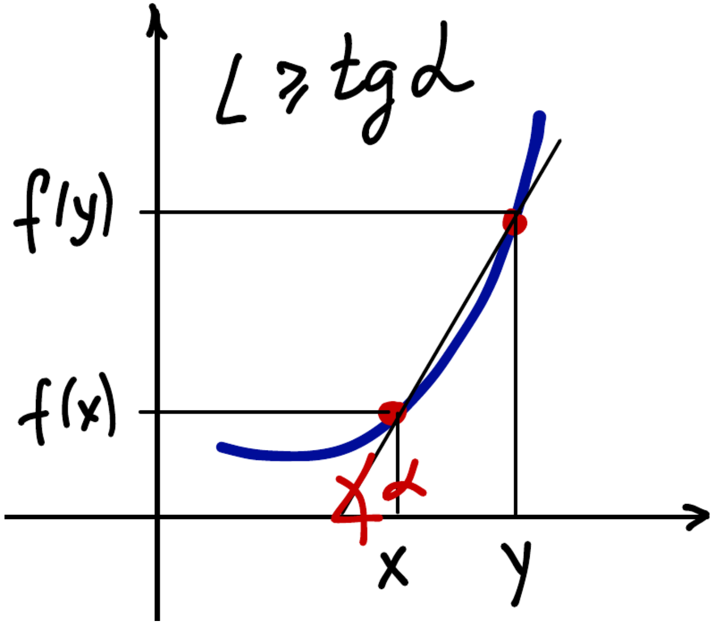

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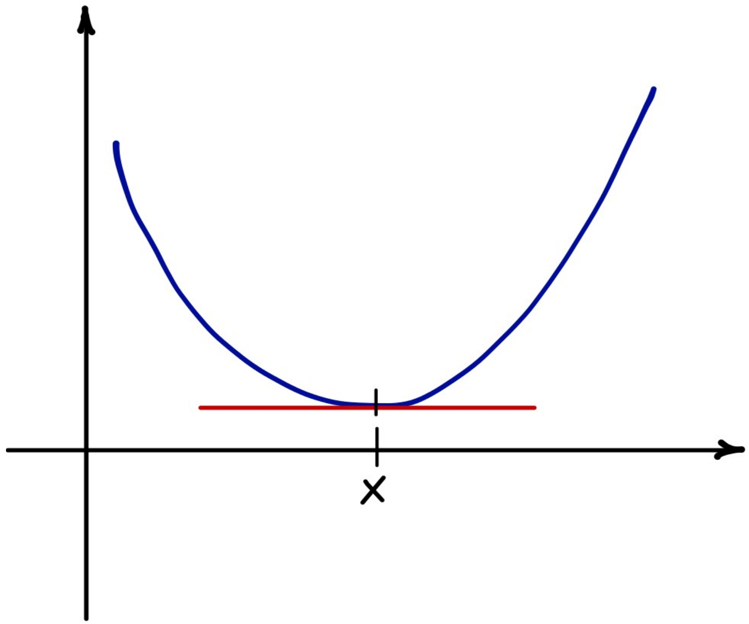
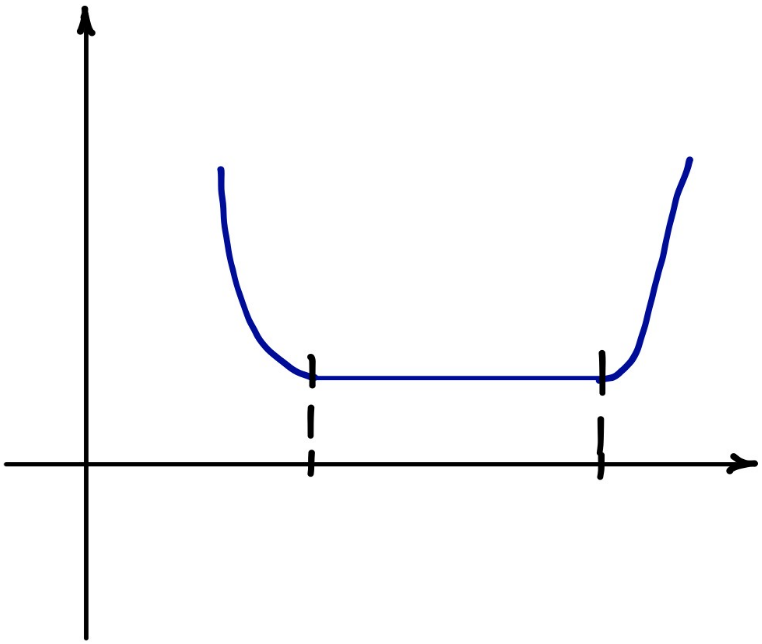
Optimality Condition

$f: \mathbb{R}^n \to \mathbb{R}$ - convex and differentiable. *x* is an optimum of *f* iff

$$\nabla f(x) = 0$$



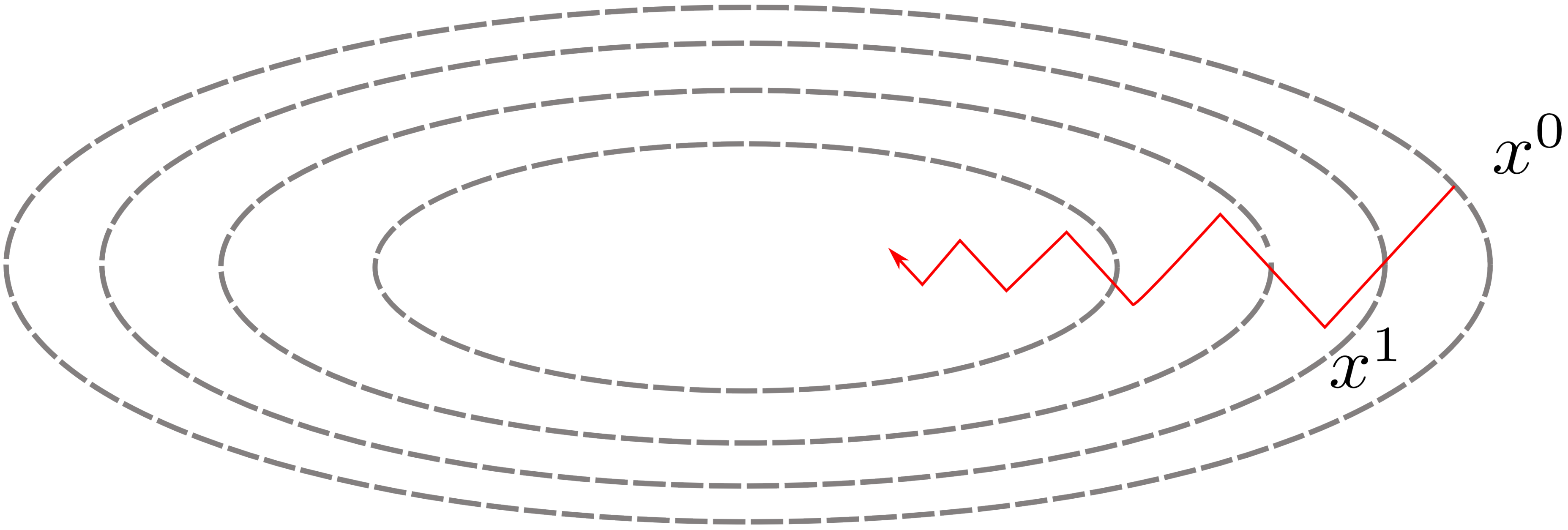


$$f - \text{convex}, \ \ D - \text{convex} \Rightarrow \arg\min_{x \in D} f(x) - \text{convex set}$$

Gradient Descent

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ - convex and differentiable.

Gradient descent: $$x^{t+1} = x^t - \alpha^t \nabla f(x^t)$$



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Total improvement of the objective: $$\epsilon := f(x^t) - f^* \leq \frac{2L\|x^0 - x^*\|}{t+4}$$

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Convergence rate: $O(\frac{L}{t})$ or $O(\frac{L}{\epsilon})$

$\epsilon = 0.1 \quad t = 10$

$\epsilon = 0.01 \quad t = 100$

Gradient Descent: Step Size Selection

Gradient descent: $$x^{t+1} = x^t - \alpha^t \nabla f(x^t)$$

Constant step size: $$a^t = \frac{1}{L}$$ Loose estimates of *L*

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Computationally intensive

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Computationally intensive

Approximate linear search: $$f(x^{t+1}) \leq f(x^t) - \frac{1}{2a^t} \|\nabla f(x^t)\|^2$$

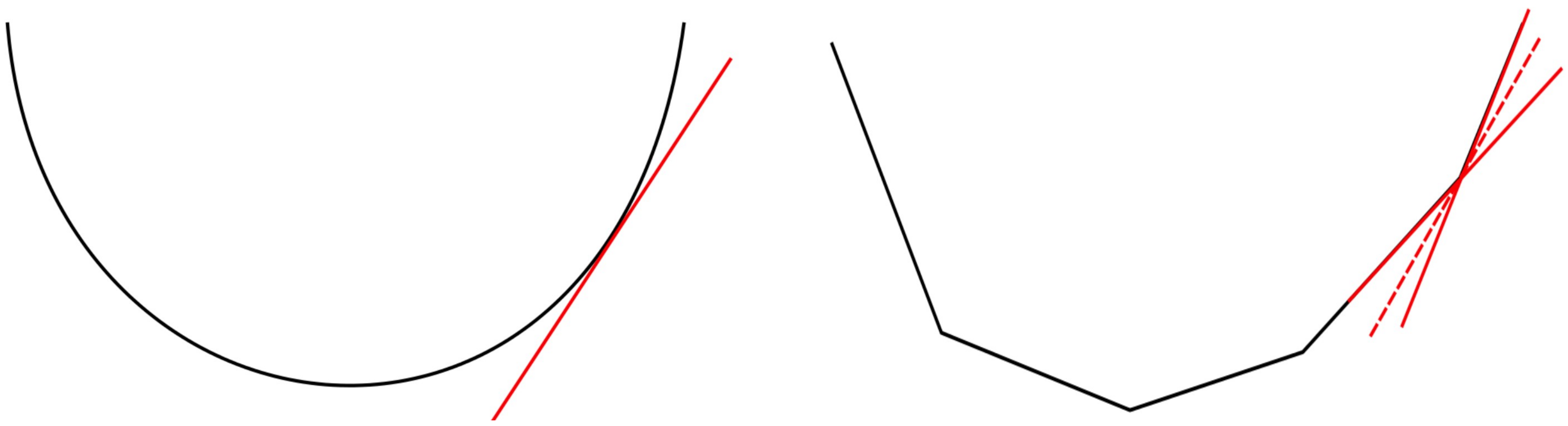
$$a^t = \frac{2a^{t-1}}{2^n}, \; n = 0, \ldots, \infty$$

Sub-Gradient: Definition

f – convex:

$$\text{Let for all } x: \ f(x) \geq f(x^0) + \langle g, x - x^0 \rangle$$

$$g - \text{subgradient of } f \text{ in } x^0 \in \operatorname{dom} f \,.$$

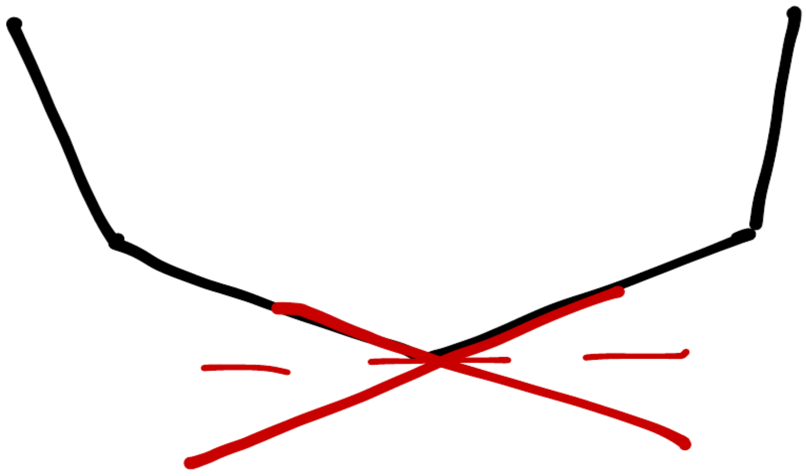


$\partial f(x) - \text{set of all subgradients in } x:$ convex, closed and bounded

f – convex, differentiable in *x*: $\partial f(x) = \{\nabla f(x)\}$

Optimality For Non-Smooth Convex Functions

Let *f* be convex, *x** - optimum of *f* iff $0 \in \partial f(x^*)$



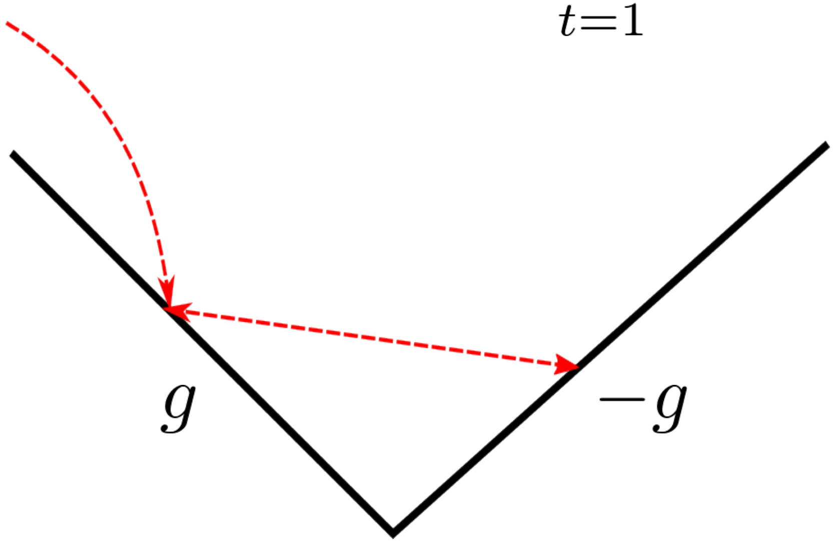
Non-zero sub-gradients can exist even in the optimum!

Sub-Gradient Method

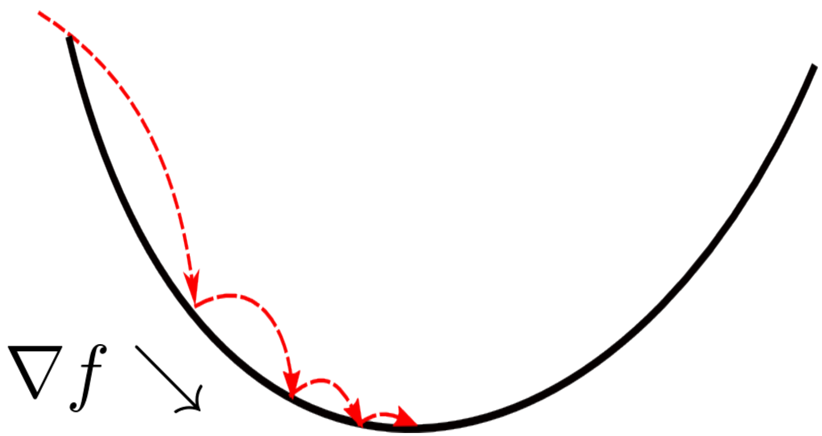
Let *f* be convex, Lipschitz continuous and $\|x^0 - x^*\| \leq R$

$$x^{t+1} = x^t - \alpha^t g(x^t)$$

$$\alpha_t > 0,\ \alpha^t \xrightarrow{t \to \infty} 0,\ \sum_{t=1}^{\infty} \alpha_t = \infty \qquad \text{e.g. } \alpha^t \propto \frac{1}{t}, \frac{1}{\sqrt{t}}$$



Constant step-size



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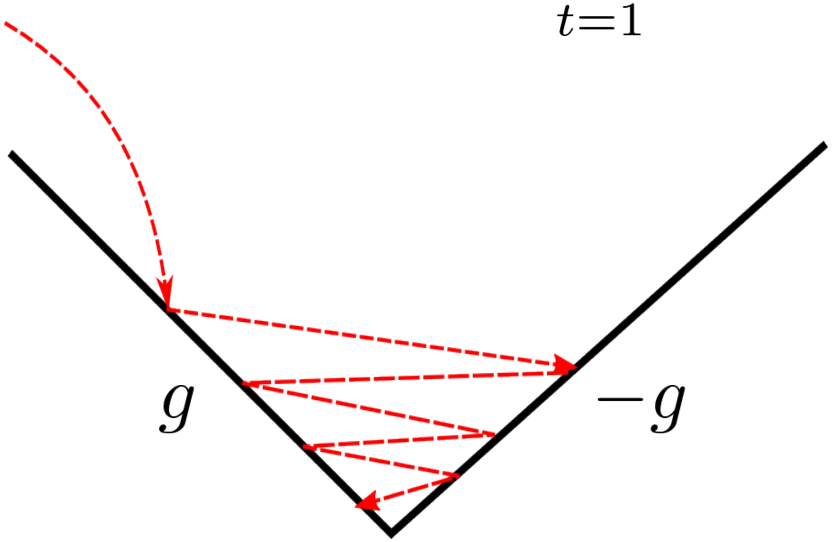
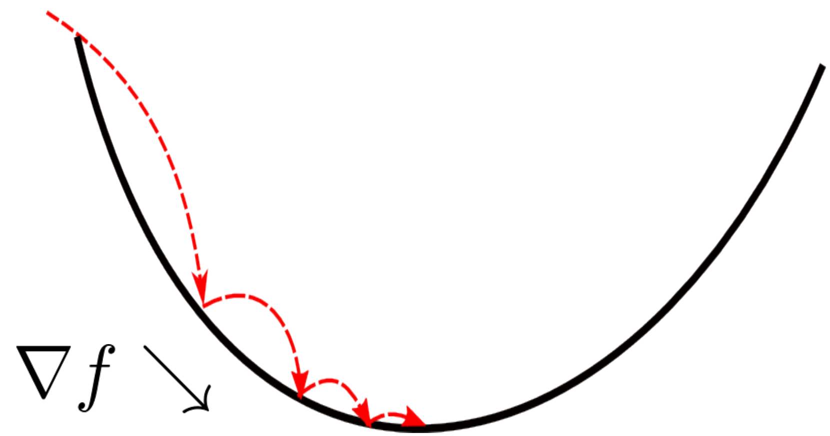


Diminishing step-size



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